

- (a) Consider a coaxial capacitor having inner radius a , outer radius b , unit length, and filled with a material having dielectric constant, ϵ_r . Compare this to a parallel-plate capacitor having plate width w , plate separation d , filled with the same dielectric, and having unit length. Assuming same energy storage, express the ratio b/a in terms of the ratio d/w for a given applied voltage, V_0 . Show complete calculations of both capacitors using Laplacian operator.
- (b) The field $B = -2a_x + 3a_y + 4a_z$ mT is present in free space. Find the vector force using $F = \oint I dL \times B$, exerted on a straight wire carrying 12 A in the a_{AB} direction, given $A(1, 1, 1)$ and $B(2, 1, 1)$.

Start your solution below

- (a) A point charge of $Q_1 = 0.35 \mu\text{C}$ is located at $r = 0$, and uniform surface charge densities are located as follows: $\rho_{s2} = 2.5 \frac{\text{mC}}{\text{m}^2}$ at $r = 1 \text{ cm}$, and $\rho_{s3} = -0.9 \frac{\text{mC}}{\text{m}^2}$ at $r = 1.0 \text{ cm}$. Determine electric flux density D at: (i) $r = 0.5 \text{ cm}$; (ii) $r = 1.5 \text{ cm}$; (iii) $r = 2.5 \text{ cm}$. (iv) What uniform surface charge density should be established at $r = 3 \text{ cm}$ to cause $D = 0$ at $r = 3.5 \text{ cm}$?
- (b) Suppose that we have two coaxial cylindrical conductors, the inner of radius a and the outer of radius b , each infinite in extent (Figure 1). We will assume a charge distribution of ρ_s on the outer surface of the inner conductor. Using symmetry and a suitable Gaussian surface of length L , mathematically formulate D everywhere.
- (c) Let us select a 50 cm length of coaxial cable having an inner radius of 1 mm and an outer radius of 4 mm . The space between conductors is assumed to be filled with air. The total charge on the inner conductor is 30 nC . Calculate (i) charge densities on each conductor, and (ii) the E and D fields.

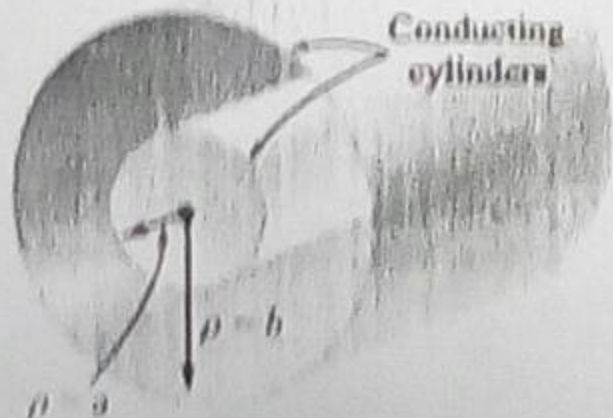


Figure 1. Two coaxial cylindrical conductors forming a coaxial cable.

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- (a) Region 1 ($x \geq 0$) is a dielectric with $\epsilon_{r1} = 2$, while region 2 ($x < 0$) has $\epsilon_{r2} = 5$. Let $E_1 = 20 \mathbf{a}_x - 10 \mathbf{a}_y + 50 \mathbf{a}_z$ V/m. (i) Find D_2 . (ii) Find the energy density in both regions.
- (b) Let the cylindrical surfaces $\rho = 4$ cm and $\rho = 9$ cm enclose two wedges of perfect dielectrics, $\epsilon_{r1} = 2$ for $0 < \varphi < \pi/2$ and $\epsilon_{r2} = 5$ for $\pi/2 < \varphi < 2\pi$. If $E_1 = \left(\frac{2000}{\rho}\right) \mathbf{a}_\rho$ V/m, draw the two wedges and find (i) E_2 ; (ii) the total electrostatic energy stored in a 1 m length of each region.

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- (a) Given the field $D = 6\rho \sin \frac{1}{2}\varphi \mathbf{a}_\rho + 1.5\rho \cos \frac{1}{2}\varphi \mathbf{a}_\varphi \text{ C/m}^2$, evaluate both sides of the divergence theorem for the region bounded by $\rho = 2$, $\varphi = 0$, $\varphi = \pi$, $z = 0$, and $z = 5$. Draw the diagram of the region as well.
- (b) Consider the portion of a sphere shown in Figure 2. The surface is specified by $r = 4$, $0 \leq \theta \leq 0.1\pi$, $0 \leq \varphi \leq 0.3\pi$, and the closed path forming its perimeter is composed of three circular arcs. We are given the field $H = 6r \sin \varphi \mathbf{a}_r + 12r \sin \theta \cos \varphi \mathbf{a}_\varphi$ and required to evaluate each side of Stokes' theorem.

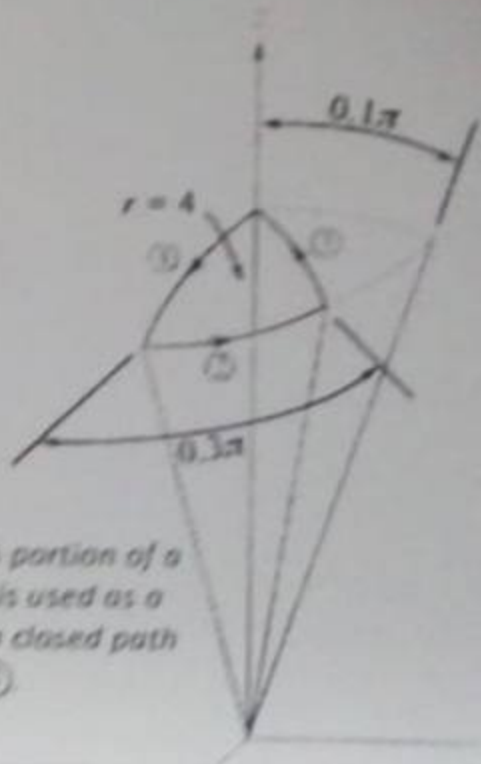


Figure 2 (b). A portion of a spherical cap is used as a surface and a closed path ① - ② - ③.

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- (a) Find magnetic field intensity H in rectangular components at $P (2, 3, 4)$ if there is a current filament on the z axis carrying 8 mA in the a_z direction. Again find H if the filament is located at $x = -1, y = 2$. Finally, find H if both filaments are present. Sketch the field in each case.
- (b) Consider a sheet of current flowing in the positive x direction defined by $K = K_x a_x \left[\frac{\text{A}}{\text{m}} \right]$ and located in the $z = 0$ plane. The return path of the current is a distant sheet with height $z = h$ and current in the opposite direction. Using symmetry and suitable Amperian loop (*diagram is must*), determine H everywhere.
- (c) Now suppose we have three current sheets, $K_1 = 2.7 a_x \frac{\text{A}}{\text{m}}$ at $y = 0.1$, $K_2 = -1.4 a_x \frac{\text{A}}{\text{m}}$ at $y = 0.15$, and $K_3 = -1.3 a_x \frac{\text{A}}{\text{m}}$ at $y = 0.25$. Determine the value of H in rectangular components everywhere and record answers in the following table.

Regions $\rightarrow$	$-\infty < y < 0.1$	$0.1 < y < 0.15$	$0.15 < y < 0.25$	$y > 0.25$
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